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Corporate ESG decoupling and R&D investment

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ABSTRACT

This study investigates whether and how firms' engagement in ESG decoupling leads to changes in R&D investment. Using a sample of U.S. listed firms from 2012 to 2023, we discover a consistent negative effect of ESG decoupling on R&D investment, indicating opposite effects from ESG brown-washing versus green-washing. Brown-washing firms exhibit a significant increase in R&D investment. Cross-sectional tests support the strategic incentive that a more pronounced positive effect observed in smaller firms, firms facing greater financial constraints and market competition, and among high-tech firms. In contrast, we find that green-washing firms experience a significant decrease in R&D investment. The decrease in R&D investment among green-washing firms is mitigated by more stringent corporate governance enforced by institutional investors, but is further amplified among firms facing greater market competition and high-tech firms, suggesting that green-washing firms with decreased R&D investment are subject to managerial opportunism. Our findings remain robust to different subsets of benchmarking normal firms and alternative measurement. In addition, we find that the capital market responds positively to ESG green-washing and negatively towards brown-washing, which implies a favorable attitude toward floated ESG disclosure from the investors. Overall, our study unveils the important role of ESG decoupling in reshaping corporate investment decision and contribute to the growing literature on ESG decoupling.

1. Introduction

Innovation has long been recognized as the key driver of corporate endogenous growth (Kogan, Papanikolaou, Seru, & Stoffman, 2017; Romer, 1994). Environmental, social, and governance (ESG) have gained prominence more recently, highlighting both financial and nonfinancial significant risks on company performance and portfolio returns (Pollman, 2019). The current view of management considers ESG and innovation as important intangible investments, because they can create assets that provide firms with a competitive edge in the long term (Gillan, Koch, & Starks, 2021; McGrath, Tsai, Venkataraman, & MacMillan, 1996). However, the management faces the challenge of making wise trade-offs in capital budgeting, particularly in terms of ESG and innovation. From a resource-based view of the firm (Wernerfelt, 1984), the exploitation of investments of various kinds and the relevant strategic activities are critical to the firms' future performance.

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Firms not only make decisions on whether to invest in ESG or not, they also deliberate on how to represent (or misrepresent) their ESG engagements. For example, in the notorious car emissions fraud broke in 2015 known as *Dieselgate*, the car company Volkswagen employed emission-cheating software on car engines to manipulate emission tests.[1] In other words, Volkswagen deceitfully pretended to be environmentally friendly, presenting low car emissions, while the actual emission levels were significant. Li and Wu (2020) find that public firms are prone to taking decoupled ESG actions without subsequently achieving real impact.

There is a burgeoning literature on the disparity between ESG disclosure and the actual performance, known as ESG decoupling (Ali Gull, Hussain, Akbar Khan, Nadeem, & Mansour Zalata, 2023; Cho, Laine, Roberts, & Rodrigue, 2015; Gull, Hussain, Khan, Khan, & Saeed, 2023; Kim & Lyon, 2015). Some companies may exaggerate ESG disclosures to appear to be trustworthy and environmentally friendly without significant investment in ESG practices (i.e., ESG green-washing) (Berrone, Fosfuri, & Gelabert, 2017; Du, 2015; Kim & Lyon, 2015; Szabo & Webster, 2020; Testa, Miroshnychenko, Barontini, & Frey, 2018; Yu, Van Luu, & Chen, 2020; Zhang, 2022). Others understate their actual ESG engagements (i.e., ESG brown-washing) due to fear of adverse reactions from stakeholders (Huang, Francoeur, & Brammer, 2022; Kim & Lyon, 2015; Testa et al., 2018). To the best of our knowledge, the literature has limited findings pertaining to the relationship between ESG decoupling and corporate investment in R&D. We aim to address this important research gap.

Based on a U.S. sample of 11,869 firm-year observations from 2012 to 2023, we examine the effect of ESG decoupling on R&D investment and explore its underlying mechanism. Specifically, we leverage an incident-based ESG performance metric from RepRisk, in conjunction with the ESG disclosure score from Bloomberg, to construct a reliable proxy that effectively captures ESG decoupling within industry clusters. Through this approach, we are able to identify variations in ESG decoupling and its impact on corporate decision-making concerning R&D investment within industries.

Our main findings indicate that ESG decoupling is associated with a decrease in firms' subsequent R&D expenditure. To illustrate it more thoroughly, we examine the R&D investment of brown-washing firms and green-washing firms separately, using normal firms (i.e., firms that do not engagement in ESG brown-washing as well as green-washing) as benchmarks. Further evidence indicates that ESG brown-washing firms show a significant increase (63.6 %) in R&D investment, compared with normal firms in the same industry. In contrast, ESG green-washing firms exhibit a notable decrease (24.2 %) in R&D investment, compared with normal firms in the same industry. These results remain after using different subsets of normal firms as benchmarks, employing alternative measurements for R&D investment and using the firm-year observations covering only the two-year periods when the focal firm changes to ESG brown-washing/green-washing in the next fiscal year, as robustness checks.

In addition, we investigate how the market responds to ESG decoupling. By using Tobin's Q as the proxy for market perception of the firm, we find that the capital market views green-washing firms as valuable investing venues, while having negative reaction toward brown-washing firms. The results imply a favorable attitude toward floated ESG disclosure from the investors, which largely aligns with the ESG investing literature (Serafeim, 2020; Cornell, 2021).

Further cross-sectional evidence reveals contrasting incentives underlying ESG brown-washing versus green-washing. The positive effect of ESG brown-washing on R&D investment is more pronounced when the firms have relatively fewer resources (both in terms of assets and financial access), face more pressure from the competitors in the product market, and have a more urgent need for technology upgrade. It reveals that ESG brown-washing firms with increased R&D investment are primarily driven by strategic incentives to excel in business operations.

In contrast, we find that the decrease in R&D investment among green-washing firms is mitigated by more stringent corporate governance enforced by institutional investors, but is further amplified among firms facing greater market competition and operating in technology-intensive industries. It supports the notion that ESG green-washing firms with decreased R&D investment are subject to managerial opportunism. The managers of ESG greenwashing firms are prone to shirking in R&D investment when they are expected to invest in corporate innovation.

The main contribution of this study is three-fold. First, we introduce a novel proxy for ESG decoupling by using an incident-based ESG performance metric from RepRisk. This metric highlights firms' realized ESG risks, making it more objective than other ratings-based metrics. Our construction is also superior to prior research in effectively capturing ESG decoupling within industry clusters. This allows us to rule out the inherent variations in ESG engagements across different industries.

Second, our study is most closely related to a strand of ESG decoupling literature. Prior researchers generally perceive ESG decoupling, especially ESG green-washing, as opportunistic and unethical (Ali Gull et al., 2023; Bothello, Ioannou, Porumb, & Zengin-Karaibrahimoglu, 2023; Gull, Hussain, Khan, Khan, & Saeed, 2023; Gull, Hussain, Khan, Mushtaq, & Orij, 2023). In contrast, ESG brown-washing is a kind of strategic endeavor against potential adverse reactions (Kim & Lyon, 2015) and is most likely to be observed among firms with higher levels of corporate legitimacy (Huang et al., 2022). Consistent with prior studies, we reveal that companies primarily engage in green-washing in ESG due to opportunistic purposes, whereas ESG brown-washing is largely driven by strategic incentives. While prior research provides a broad understanding, our study takes a more focused approach by examining the specific impact of ESG decoupling on R&D investment through rigorous empirical research. Our study offers important research and managerial implications, suggesting that ESG decoupling could have far-reaching impacts on seemingly unrelated corporate decisions and firm value.

Third, our findings also contribute to the literature on the determinants of corporate R&D investment. Existing literature implies that ESG-related corporate activities can have significant impacts on corporate innovation. Broadstock, Matousek, Meyer and

[1] The Wall Street Journal, SEC Charges Volkswagen With Defrauding U.S. Bond Investors; SEC says German auto maker raised billions of dollars while making deceptive claims about environmental impact of 'clean diesel' fleet, 15 March 2019.

Tzeremes (2020) support that ESG adoption enhances firms' abilities to pursue innovation and eventually have a positive impact on firm performance. Tsang, Wang, Liu and Yu (2021) find that integrating ESG into executive compensation effectively spur corporate innovation. We instead focus on a less-explored perspective of ESG decoupling in understanding variations in firm-level R&D investment. Our study adds to the understanding of the important factors that both spur and hinder corporate innovation input.

The remainder of this paper is organized as follows. We provide a thorough summary of related literature, and develop research hypotheses for empirical tests in Literature Review and Hypothesis Development. Data, variables and regression models are explained in Data and Methodology, while the empirical results and analyses are included in Empirical Results. Finally, we summarize in Conclusion.

2. Literature review and hypothesis development

Many researchers analyze ESG decoupling from the green-washing perspective, where firms engage in symbolic communication to misrepresent their ESG efforts without undertaking substantive actions (Ali Gull et al., 2023; Bothello et al., 2023; Gull, Hussain, Khan, Khan, & Saeed, 2023; Gull, Hussain, Khan, Mushtaq, & Orij, 2023). In fact, when we examine the gap between ESG disclosure and performance, it can go both ways: one is ESG green-washing, where firms exaggerate ESG engagements; the other is ESG brown-washing, where firms understate the actual ESG engagements.

ESG green-washing normally reflects firms' unethical and opportunistic behaviors. Kim and Lyon (2015) imply that greenwashing emerges when being green is not a win–win proposition and the firms have incentives to exaggerate in information disclosure, upon the tension between the demands of shareholders and pressures from secondary stakeholders. Zhang (2022) provides evidence on that the firms' ESG greenwashing is motivated by financial constraints. CEO power (Gull, Hussain, Khan, Mushtaq, & Orij, 2023) and interfirm affiliation (Bothello et al., 2023) can also contribute to the ESG disclosure in a more opportunistic manner.

In contrast, Kim and Lyon (2015) regard ESG brown-washing as a rational strategy against potential adverse reactions from stakeholders on ESG investment. This perception is largely in line with a body of literature on unethical pro-organizational behavior (Mishra, Ghosh, & Sharma, 2021) which demonstrates that the set of actions is unethical but seeks to further the interests of the organizations. Consistently, Huang et al. (2022) find that ESG brown-washing is most likely to be observed among firms with higher levels of corporate legitimacy. However, ESG brown-washing is also found to be harmful to stakeholders (Huang et al., 2022; Testa et al., 2018).

In this regard, we conjecture that firms engaging in ESG brown-washing strategically underreport their ESG engagement to avoid accusations of making extravagant investments. As a result, they are able to increase investment in corporate innovation. On the contrary, firms engaging in ESG green-washing, out of opportunistic purposes, tend to decrease R&D investment.

Hereby Hypotheses 1 and 2 are as follows.

H1: *Firms that engage in ESG brown-washing increase the subsequent investment in R&D.*

H2: *Firms that engage in ESG green-washing decrease the subsequent investment in R&D.*

3. Data and methodology

3.1. Data and sample

In order to construct our research sample, we integrate the incident-based ESG performance data from RepRisk, ESG disclosure score data from Bloomberg, and financial and corporate governance data from Compustat and Boardex. Through the merging process and the exclusion of observations with missing data, our final dataset comprises 11,869 firm-year observations of 1,995 U.S. listed firms spanning the period from 2012[2] to 2023.

Prior studies on ESG decoupling (or ESG greenwashing), for example, Yu et al. (2020) and Gull, Hussain, Khan, Khan, & Saeed (2023), use Refinitive (former Thomson Reuters Asset4) ESG performance score as a proxy for ESG performance. In our study, to better capture the gap between ESG disclosure and actual performance, we rely on an incident-based measure of ESG performance from RepRisk, which is considered more objective than other commercial ratings and reflects realized outcomes in the ex-post (Li & Wu, 2020).

3.2. Empirical methodology

We use the following model specification to examine how firms' engagement in ESG decoupling affects R&D investment:

$$R\&D_{i,t+1} = \alpha + \beta_1 Decouple_{i,t} + \beta_2 Controls_{i,t} + \delta_s + \delta_t + \varepsilon_{i,t+1} \quad (1)$$

The dependent variable, *R&D*, signifies the ratio of R&D expenditure to total assets for a firm in a given year. Due to a large number of firms withholding their R&D expenditure in reporting (Koh & Reeb, 2015), we address this issue by filling missing R&D with the industry average R&D value. Additionally, we also set missing value to zero as robustness check.

Our explanatory variable of interest, *Decouple*, takes on a value of one if the firm has both above-median ESG and above-median RRI

[2] The starting year with the available Bloomberg ESG disclosure score is 2012.

within the three-digit SIC cluster for the year; it takes the value of minus one if the firm has both below-median ESG and below-median RRI within the three-digit SIC cluster for the year; zero otherwise. Our construction is similar to that in Yu et al. (2020) as well as in Gull, Hussain, Khan, Khan, & Saeed (2023), except that we use the annual average of *Peak_RRI* from RepRisk as the proxy for ESG performance. To show the distinct impact from ESG brown-washing versus ESG green-washing separately, we further construct the *Brown* and *Green* dummy variables. The variable Brown is a dummy variable that equals one if *Decouple* is minus one, and zero otherwise, serving as a proxy for firms engaging in ESG brown-washing. Similarly, the variable *Green* is a dummy variable with a value of one if *Decouple* is one, and zero otherwise, acting as a proxy for firms involved in ESG green-washing.

Consistent with prior research, we incorporate a set of control variables on firm-level characteristics, that may have important explanatory power on R&D expenditure. Among these variables, leverage (*Lev*), measured as debt over asset, plays a crucial role in determining the extent to which funds can be allocated to R&D investment (David et al., 2008). We introduce profitability (*ROE*, return on equity), considering that sustained R&D investment necessitates long-term financial support, which is contingent upon the firm's profitability (Coad and Rao, 2010; Lai et al., 2015).

Our model also accounts for market dynamics, including market-to-book ratio (*MB*), and sales growth ratio (*Growth*). These factors collectively capture the market's assessment of growth opportunities for the firm and influence its decisions regarding R&D spending (Peters and Taylor, 2017; Liang, 2012). The size (*Size*) and age (*Age*) of the firm are also considered, given their recognized importance in influencing innovation and R&D investment, as respectively indicated by Fishman and Rob (1999), and Coad et al. (2016) as well as García-Quevedo et al. (2014).

In the realm of corporate governance factors, we first incorporate board size (*Boardsize*). Larger boards are often associated with challenges in reaching a consensus, particularly for high-stakes decisions such as R&D investments (Zona, 2016). The proportion of outside directors on the board (*Outratio*) is also considered for its independent role in facilitating firm R&D expenditure (Yoo and Sung, 2015; Kang et al., 2018). Furthermore, the presence of female directors on the board (*Femratio*) is a potential avenue for improving governance, managing risk, and enhancing the performance of R&D investment, as suggested by Chen et al. (2016; 2018). Finally, it's worth noting that previous studies have highlighted the impact of CEO duality (*Dual*) and CEO tenure (*Tenure*) on R&D decisions (Zona, 2016).

Moreover, our model specification mainly includes industry fixed effect (δs) and year fixed effect (δt) to reflect within-industry across-firm variations of the impact of ESG decoupling on R&D investment.

Appendix provides detailed definition of all the variables.

Table 1
Sample distribution. This table presents our research sample distribution by year as well as by industry.

Panel A- Distribution by year

Year	Brown	Normal	Green	Total
2012	181	496	227	904
2013	203	486	260	949
2014	234	432	293	959
2015	308	383	315	1,006
2016	288	404	297	989
2017	320	436	337	1,093
2018	340	447	352	1,139
2019	321	462	334	1,117
2020	307	418	314	1,039
2021	295	392	293	980
2022	336	422	340	1,098
2023	180	241	175	596
	3,313	5,019	3,537	11,869

Panel B – Distribution by industry

Industry	Brown	Normal	Green	Total
Agriculture, Forestry and Fishing	9	14	7	30
Mining & Construction	297	313	299	909
Manufacturing	1,373	1,976	1,431	4,780
Transportation, Communications, Electric, Gas and Sanitary service	249	350	276	875
Trade	394	507	403	1,304
Finance, Insurance and Real Estate	423	616	461	1,500
Services	451	897	535	1,883
Non-Operating Establishments and Others	117	346	125	588
Total	3,313	5,019	3,537	11,869
High-tech[1]	1,417	2,256	1,595	5,268
Non-high-tech	1,896	2,763	1,942	6,601

[1]We define high-tech firms if the firm is in the "Computer", "Biotech/Medical", "Electronics", and "Telecom" industries which follows Puri and Zarutskie (2012)'s detailed SIC classification.

3.3. Summary statistics

Panels A and B of Table 1 illustrate the distribution of sample observations by year and by industry. According to our identification strategy of *Brown*, *Normal*, and *Green* firms, the distribution should be balanced across years and industries. The panels both show balanced results but also reflect notable variations.

In Panel A, it indicates that the number of brown-washing firms is comparable to that of green-washing firms across years, but both are lower than the count of normal firms. The observations in 2023 are smaller than in previous years due to missing values in Boardex as of the fiscal year 2023. Analyzing the temporal trend, brown-washing firms display an initial increase followed by a decline, peaking at 340 in 2018 and reaching a minimum of 181 in 2012. Similarly, the quantity of green-washing firms follows a comparable increasing–decreasing pattern, reaching its peak at 352 in 2018. In contrast, normal firms exhibit a generally declining trend, signifying an increasing corporate focus on ESG and its transformation from optional to essential information.

Panel B presents the disparities in the industry distribution of the three types of firms. Across all industries, except for the Agriculture and Non-operating establishments sectors, the number of green-washing firms surpasses that of brown-washing firms, indicating a corporate inclination towards leveraging ESG disclosure. Notably, the industry with the highest prevalence of ESG green-washing firms, outnumbering brown-washing firms, is the Services sector. Further categorizing industries into high-tech and non-high-tech, the analysis reveals that, in both groups, the quantity of green-washing companies remains higher than that of brown-washing firms.

Table 2 displays descriptive statistics for the primary variables. The mean of *R&D*, with missing values imputed using the industry average for each year, stands at 0.066 in our research sample. In contrast, the alternative measure of *R&D*, denoted as *R&D'*, which assigns zeros for missing R&D values, has an average of 0.029, lower than that of *R&D*. Additionally, ESG decoupling (*Decouple*) demonstrates an average value of 0.019 with a standard deviation of 0.759. This implies that we identify more firms engaging in ESG green-washing than brown-washing.

In Table 3, the correlation matrix is presented to assess potential multicollinearity among variables. Notably, *R&D* and *R&D'* are strongly positively related, with a correlation coefficient of 0.592. As for *Decouple*, the outcome variable, its correlation with *R&D* as well as *R&D'* in the same year is inconclusive, with the coefficient with *R&D* being significantly negative, while the coefficient with *R&D'* is significantly positive.

4. Empirical results

4.1. Main regression results: ESG decoupling and R&D expenditure

First, we visually present the raw relationship between ESG decoupling and R&D investment with binned scatterplots, which are useful for detecting scientific apophenia and identifying outliers and distributional concerns (Starr and Goldfarb, 2020). Fig. 1 presents a scatterplot illustrating the variation in R&D expenditure from 2013 to 2023 for firms categorized as brown, normal, or green in the previous fiscal year. Contrasting the differences among these three types of firms, it is evident that the R&D investment of brown-washing firms consistently surpasses that of the other two groups.

In Fig. 2, we present a scatterplot illustrating the relationship between R&D expenditure and Bloomberg ESG disclosure scores for firms categorized as *Brown*, *Normal*, or *Green* in the previous fiscal year. The negative correlation between R&D expenditure and ESG scores for all firms is evident, as the plot reveals a consistent decline in R&D expenditure with the increase in ESG disclosure scores. However, the rate of decrease varies among the three types of companies. Specifically, for brown-washing firms, the plot indicates a pronounced reduction in R&D expenditure with the ascent of Bloomberg ESG disclosure scores. In contrast, for green-washing firms, the increase of ESG disclosure scores does not lead to a significant decrease in R&D expenditure. This disparity highlights that, in order to undertake substantial actions supporting ESG development, brown-washing firms find it necessary to substantially cut investments in R&D to allocate resources for ESG initiatives. On the other hand, green-washing firms, which tend to superficially engaged in ESG efforts, do not need to sacrifice a substantial portion of their R&D investments.

Fig. 3 illustrates the relationship between R&D expenditure and RepRisk RRI for firms categorized as *Brown*, *Normal*, or *Green* in the previous fiscal year. It suggests that ESG brown-washing firms with lower ESG risk exposure have higher level of R&D investment in the next fiscal year. In contrast, ESG green-washing firms with higher ESG risk exposure tend to have lower level of R&D investment in the next fiscal year. For the rest of the firms, R&D investment and RepRisk RRI do not show a significant relationship.

We proceed to conduct the regression model as outlined in Eq.(1) and present the primary regression outcomes in Table 4. In column (1), controlling for firm-level financial factors, the coefficient of the explanatory variable, *Decouple*, is −0.036, displaying statistical significance at the 1 % level. This finding suggests that ESG decoupling is associated with a significant decrease in firms' R&D investment in the next fiscal year. We introduce corporate governance variables in the control set in column (2) and find that the negative relationship between ESG decoupling and firms' R&D investment remains significant. We additionally add state fixed effect in the regression of column (3) to account for heterogeneity in the state level. The negative impact of ESG decoupling on R&D investment remains significant in the regression.

In columns (4) and (5), we distinctly examine ESG brown-washing and green-washing firms. Column (4) shows the results using the

Table 2
Descriptive statistics. This table presents the descriptive statistics for key variables. See definitions for all variables in Appendix.

VarName	Obs	Mean	SD	Min	Median	Max
R&D	11,869	0.066	0.348	0.000	0.012	17.365
R&D′	11,869	0.029	0.082	0.000	0.000	2.106
Decouple	11,869	0.019	0.759	−1.000	0.000	1.000
Size	11,869	8.029	1.782	3.168	7.977	12.727
Lev	11,869	0.306	0.225	0.000	0.286	1.118
ROE	11,869	0.177	0.740	−3.725	0.178	3.592
Q	11,869	1.907	1.690	0.132	1.367	10.576
MB	11,869	1.885	1.698	0.191	1.341	10.613
Age	11,869	3.110	0.748	0.693	3.178	4.290
Growth	11,869	0.108	0.310	−0.593	0.061	2.153
Boardsize	11,869	2.992	0.406	1.946	2.996	3.951
Outratio	11,869	0.211	0.569	0.000	0.000	3.069
Femratio	11,869	1.291	1.004	0.000	1.170	4.235
Dual	11,869	0.356	0.479	0.000	0.000	1.000
Tenure	11,869	5.877	4.981	1.000	4.000	26.000

independent variable *Brown*, where the coefficient is 0.042 and statistically significant at the 1 % level. This implies that firms engaged in ESG brown-washing practices tend to exhibit 63.6 %[3] higher R&D investment, when compared with their normal counterparts in the same industry. Conversely, the coefficient of the *Green* variable is −0.016 shown in column (5) and statistically significant at the 1 % level, indicating that firms involved in ESG green-washing practices tend to have 24.2 %[4] lower R&D expenditure, compared with normal firms in the same industry. The results reported in Table 4 are consistent with the trend in Fig. 1 and are also align with the expectations in Hypothesis 1 and Hypothesis 2.

4.2. The market perception of ESG decoupling

Next, we investigate the evolving market perception of ESG decoupling. Specifically, Tobin's Q is employed to assess market perception, as it reflects the market perception of the firms' future investment opportunity and value creation. In columns (1) and (2) of Table 5, Tobin's Q serves as the dependent variable, and *Decouple* as the independent variable. The coefficients of *Decouple* are all positively signed and statistically significant at the 1 % level, with values of 0.179 and 0.172, respectively.

Taking a step further, we explore whether the market responds differently to ESG brown-washing firms compared to green-washing firms. In columns (3) and (4), it is noteworthy that the capital market exhibits a positive response only to ESG green-washing firms, showing a negative response to brown-washing firms. This is evidenced by the positively significant coefficient of the *Green* variable (0.161, significant at the 1 % level), while the coefficient of the *Brown* variable is significantly negative (−0.133, significant at the 5 % level). That is to say, comparing with normal firms, market responds positively to ESG green-washing firms and negatively to ESG brown-washing firms.

The empirical results suggest that investors generally consider firms' ESG disclosure as a crucial factor in valuing them. However, they may overlook the actual practices of these firms. This corroborates prior findings that public sentiment, which can be influenced by corporate disclosure, marketing campaigns, and other factors, leads to the value of corporate sustainability deviating from its actual performance (Serafeim, 2020; Cornell, 2021).

These findings also support the notion that ESG decoupling serves as an effective disguising tool. The positive market response to ESG green-washing and the negative response to brown-washing indicate that investors genuinely believe in the disclosed ESG commitment. Consequently, managers can effectively utilize ESG green-washing to shirk their responsibilities and employ brown-washing strategically to allocate resources towards corporate innovation.

4.3. Robustness check

To validate our main findings, we employ a set of robustness checks including using different subset samples with normal firms conditional on the ESG and RRI levels as benchmark and using alternative measure for the dependent variable *R&D*. Panel A of Table 6 shows the regression results using different subsets of normal firms conditional on the ESG and RRI levels. In column (1), we compare ESG brown-washing firms with normal firms with below-median Bloomberg ESG scores, where the coefficient for *Brown* is 0.043 and statistically significant at the 1 % level. Moving to column (2), we again discern brown-washing firms from normal ones with below-median RRI, with the coefficient of *Brown* at 0.053 and statistically significant at the 1 % level. The similar approach is applied in column (3) and (4) for the difference between green-washing and normal ones, and the coefficients of *Green* variable are consistently negative and strongly significant at 1 % level, with values of −0.015 and −0.027, respectively. The results in Panel A support the robustness of our finding in Table 4.

[3] =0.042*1/0.066.
[4] =-0.016*1/0.066.

Table 3
Correlation matrix. This table presents Spearman correlation among key variables used in the main regressions. See definitions for all variables in Appendix.

	R&D	R&D′	Decouple	Size	Lev	ROE	Q	MB	Age	Growth	Boardsize	Outratio	Femratio	Dual	Tenure
R&D	1														
R&D′	0.592***	1													
Decouple	−0.134***	0.024***	1												
Size	−0.353***	−0.151***	0.476***	1											
Lev	−0.233***	−0.190***	0.078***	0.264***	1										
ROE	−0.125***	−0.106***	0.143***	0.254***	0.157***	1									
Q	0.245***	0.361***	0.024***	−0.136***	−0.025***	0.183***	1								
MB	0.250***	0.367***	0.020**	−0.149***	−0.026***	0.177***	0.997***	1							
Age	−0.107***	−0.009	0.196***	0.327***	−0.006	0.166***	−0.073***	−0.080***	1						
Growth	0.045***	0.052***	−0.063***	−0.031***	−0.023**	0.094***	0.222***	0.223***	−0.162***	1					
Boardsize	−0.160***	−0.034***	0.293***	0.444***	0.072***	0.149***	−0.083***	−0.090***	0.337***	−0.089***	1				
Outratio	−0.048***	0.001	0.018*	0.030***	−0.073***	0.054***	0.000	−0.003	0.195***	−0.045***	0.080***	1			
Femratio	−0.143***	−0.018*	0.267***	0.420***	0.118***	0.116***	0.022**	0.020**	0.291***	−0.036***	0.463***	−0.045***	1		
Dual	0.028***	0.041***	0.024***	−0.004	−0.017*	0.039***	0.018**	0.017*	0.069***	−0.004	−0.049***	0.068***	−0.046***	1	
Tenure	0.021**	−0.013	−0.040***	−0.027***	−0.050***	0.017*	0.041***	0.040***	0.127***	−0.004	−0.025***	0.130***	−0.054***	0.186***	1

***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

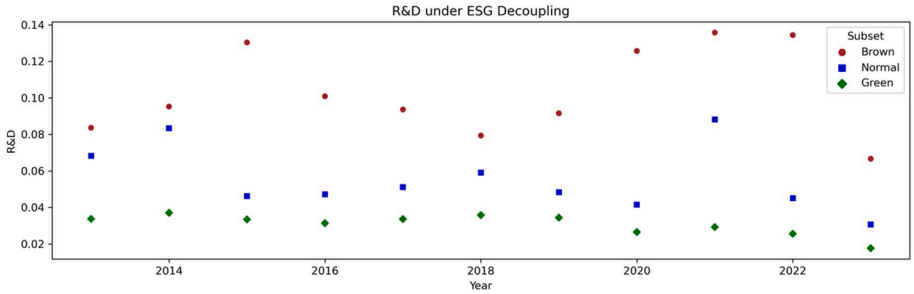

Fig. 1. R&D Expenditure under ESG Decoupling. This figure presents a scatterplot illustrating how R&D expenditure changes from 2013 to 2023 for firms categorized as brown, normal, or green(The firm is defined as brown if *Brown* = 1 (i.e., *Decouple* = -1); as normal if *Decouple* = 0; as green is *Green* = 1 (i.e., *Decouple* = 1).) in the previous fiscal year (The year of 2012 is missing because of lack of Bloomberg ESG data before 2012.).

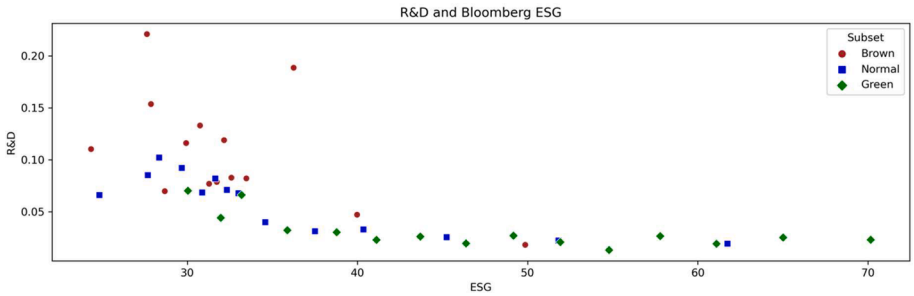

Fig. 2. R&D Expenditure and Bloomberg ESG. This figure presents a scatterplot illustrating how R&D expenditure responds to Bloomberg ESG for firms categorized as brown, normal, or green in the previous fiscal year.

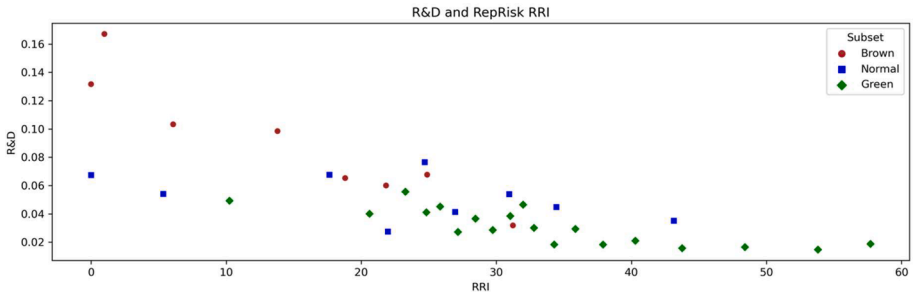

Fig. 3. R&D Expenditure and RepRisk RRI. This figure presents a scatterplot illustrating how R&D expenditure responds to RepRisk RRI for firms categorized as brown, normal, or green in the previous fiscal year.

In Panel B, we employ an alternative measure for the dependent variable, denoted as $R\&D'$, representing R&D expenditure (with missing values filled by 0) over total assets. In column (1), the coefficient for *Brown* is significantly negative with a value of -0.005. Upon changing the independent variable to *Brown*, the coefficient becomes 0.008 in column (2). Moving to column (3), the coefficient for *Green* is -0.003. The outcomes in Panel B also affirm the stability of our main findings.

In Panel C, we confine our analysis to the firm-year observations covering only the two-year periods when the focal firm changes to ESG brown-washing/green-washing in the next fiscal year. We report ESG brown-washing, i.e., the change of *Brown* from 0 in t to 1 in $t + 1$, in columns (1) and (2), and ESG green-washing, i.e., the change of *Green* in t from 0 to 1 in $t + 1$, in columns (3) and (4). The results show that when the firm changes to ESG brown-washing, the firm's R&D investment in the next fiscal year tends to increase. However, we do not observe a significant decrease in R&D investment right after the firms changes to ESG green-washing. This indicates that the decreasing pressure on R&D investment from ESG green-washing does not arise at the time of the change, but rather manifests in a more gradual manner.

Table 4
Baseline regression of ESG decoupling and R&D expenditure. This table shows the baseline regression results of ESG decoupling on firm's R&D expenditure using the following specification: $R\&D_{i,t+1} = \alpha + \beta_1 Decouple_{i,t} + \beta_2 Controls_{i,t} + IndustryFE + YearFE + \varepsilon$. *R&D* denotes R&D expenditure (with missing values filled by industry average for each year) over total assets. Columns (1) and (2) uses *Decouple* as the explanatory variable and regress in the full sample; column (3) uses *Brown* as the explanatory variable to identify brown firms versus normal firms, while column (4) uses *Green* to identify green firms versus normal firms. All continuous variables are winsorized at the upper and lower 1 %. See definitions for all variables in Appendix.

	(1)	(2)	(3)	(4)	(5)
	Full sample	Full sample	Full sample	Brown versus normal	Green versus normal
	$R\&D_{i,t+1}$	$R\&D_{i,t+1}$	$R\&D_{i,t+1}$	$R\&D_{i,t+1}$	$R\&D_{i,t+1}$
$Decouple_{i,t}$	−0.036***	−0.027***	−0.023***		
	(−4.766)	(−4.746)	(−4.082)		
$Brown_{i,t}$				0.042***	
				(3.744)	
$Green_{i,t}$					−0.016***
					(−3.205)
$Age_{i,t}$	−0.003	0.002	−0.003	0.005	−0.004
	(−0.603)	(0.294)	(−0.495)	(0.662)	(−0.841)
$MB_{i,t}$	0.007*	0.007*	0.007*	0.005	0.010***
	(1.758)	(1.818)	(1.787)	(0.981)	(3.619)
$Lev_{i,t}$	−0.079***	−0.075***	−0.073***	−0.082***	−0.058***
	(−3.447)	(−3.404)	(−3.383)	(−2.851)	(−3.536)
$ROE_{i,t}$	−0.009*	−0.009*	−0.008	−0.011	−0.009**
	(−1.673)	(−1.657)	(−1.502)	(−1.309)	(−2.212)
$Growth_{i,t}$	−0.022*	−0.024**	−0.027**	−0.032**	−0.019
	(−1.951)	(−2.089)	(−2.239)	(−2.188)	(−1.428)
$Boardsize_{i,t}$		−0.016	−0.004	−0.024	−0.007
		(−1.024)	(−0.346)	(−1.095)	(−0.755)
$Outratio_{i,t}$		0.025	0.026	0.034	0.014
		(1.286)	(1.318)	(1.329)	(1.211)
$Femratio_{i,t}$		−0.021**	−0.021**	−0.028**	−0.015***
		(−2.339)	(−2.334)	(−2.058)	(−3.217)
$Dual_{i,t}$		0.013	0.009	0.017	−0.003
		(0.861)	(0.571)	(0.786)	(−0.589)
$Tenure_{i,t}$		−0.001	−0.001	−0.002	−0.001
		(−0.897)	(−0.651)	(−1.207)	(−0.636)
_cons	0.087***	0.141***	0.119***	0.159***	0.110***
	(5.144)	(3.478)	(3.448)	(2.949)	(3.943)
Industry FE	YES	YES	YES	YES	YES
State FE	NO	NO	YES	NO	NO
Year FE	YES	YES	YES	YES	YES
N	8,354	8,354	8,354	5,772	6,125
Adj. R^2	0.060	0.066	0.115	0.068	0.068
F	8.439	5.243	6.238	2.954	7.408

t-statistics with standard errors clustered in the firm level are in parentheses. ***, ** and * indicate statistical significance at the 1 %, 5 %, and 10 % levels, respectively.

4.4. Cross-sectional analysis

4.4.1. ESG brown-washing firms versus normal firms

The results presented in Table 7 offer insights into the cross-sectional variation in R&D expenditure (both *R&D* and *R&D'*) among ESG brown-washing firms. In columns (1) and (2), the coefficients of the interaction term between *Brown* and *Big_firm* (a dummy variable indicating whether the firm size is big or not) are consistently negative and statistically significant at the 5 % level, with values of −0.035 and −0.009, respectively. These findings imply that the smaller firms are more inclined to strategically understate corporate ESG engagement with brown-washing and subsequently increase its R&D investment. This can be explained by the fact that smaller firms are more resource-constrained, making them more likely to behave more strategically.

We observe a heightened positive effect for firms facing greater financial constraints in columns (3) and (4). The positive and significant coefficients of the interaction term between *Brown* and *WW_high* (a dummy variable indicating whether WW index is higher than median or not) imply that ESG brown-washing firms under financial constraints (or with a high WW index), are able to allocate more funds to R&D investment, in order to pursue legitimacy in terms of corporate innovation in the long term.

We also observe heterogeneity from the perspective of market competition. Columns (5) and (6) reveal positive and significant coefficients for the interaction term between *Brown* and *HHI_low* (a dummy variable indicating whether Herfindahl index is lower than median or not), It suggests that firms confronted with intensified competition (or with a low HHI), prioritize increased investments in R&D to enhance their competitiveness in the product market.

Finally, we find that the positive effect is more pronounced in high-tech ESG brown-washing firms. In columns (7) and (8), the positive coefficients of the interaction term between *Brown* and *Hightech* (a dummy variable indicating whether the firm is high-tech

Table 5
The market perception of ESG decoupling. This table shows the regression results of ESG decoupling on Tobin's Q using the following specification: $Q_{i,t+1} = \alpha + \beta_1 Decouple_{i,t} + \beta_2 Controls_{i,t} + IndustryFE + YearFE + \varepsilon$. Columns (1) and (2) use *Decouple* as the explanatory variable and regress in the full sample; column (3) uses *Brown* as the explanatory variable to identify brown firms versus normal firms, while column (4) uses *Green* to identify green firms versus normal firms. All continuous variables are winsorized at the upper and lower 1 %. See definitions for all variables in Appendix.

	(1)	(2)	(3)	(4)
	Full sample	Full sample	Brown versus Normal	Green versus Normal
	$Q_{i,t+1}$	$Q_{i,t+1}$	$Q_{i,t+1}$	$Q_{i,t+1}$
$Decouple_{i,t}$	0.179***	0.172***		
	(4.701)	(4.526)		
$Brown_{i,t}$			−0.133**	
			(−2.426)	
$Green_{i,t}$				0.161***
				(2.607)
$Age_{i,t}$	−0.033	−0.073	−0.113*	−0.036
	(−0.617)	(−1.347)	(−1.890)	(−0.603)
$Lev_{i,t}$	0.627**	0.644**	0.690**	0.597**
	(2.468)	(2.531)	(2.510)	(2.038)
$ROE_{i,t}$	0.042	0.046	0.068	0.030
	(1.015)	(1.095)	(1.463)	(0.642)
$Growth_{i,t}$	0.609***	0.617***	0.559***	0.748***
	(6.452)	(6.609)	(5.708)	(5.984)
$Size_{i,t}$	−0.153***	−0.164***	−0.183***	−0.127***
	(−5.359)	(−5.414)	(−4.822)	(−4.127)
$Boardsize_{i,t}$		−0.104	0.025	−0.182
		(−1.046)	(0.240)	(−1.617)
$Outratio_{i,t}$		0.008	0.073	−0.037
		(0.150)	(1.229)	(−0.728)
$Femratio_{i,t}$		0.137***	0.152***	0.142***
		(3.217)	(3.115)	(2.984)
$Dual_{i,t}$		0.020	0.056	−0.015
		(0.264)	(0.628)	(−0.178)
$Tenure_{i,t}$		0.017*	0.016*	0.021**
		(1.948)	(1.926)	(2.013)
_cons	2.994***	3.236***	3.045***	3.051***
	(12.378)	(10.159)	(8.557)	(8.741)
Industry FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
N	8,354	8,354	5,772	6,125
Adj. R^2	0.278	0.282	0.291	0.305
F	13.795	9.078	6.947	6.709

t-statistics with standard errors clustered in the firm level are in parentheses. ***, ** and * indicate statistical significance at the 1 %, 5 %, and 10 % levels, respectively.

firm or not) imply that high-tech firms, due to more urgent need in technology upgrade, place a greater emphasis on R&D investment.

In sum, the results from Table 7 show that the positive effect of ESG brown-washing on R&D investment is more pronounced when the firms have relatively fewer resources (both in terms of assets and financial access), face more pressure from the competitors in the product market, and have a more urgent need for technology upgrade. It reveals that ESG brown-washing firms with increased R&D investment are primarily driven by strategic incentives to excel in business operations.

4.4.2. ESG green-washing firms versus normal firms

We next explore the cross-sectional variation in R&D expenditure (both *R&D* and *R&D'*) among ESG green-washing firms in Table 8. First, we do not find significant difference between big firms versus small firms in columns (1) and (2). It suggests that the negative effect of ESG greenwashing is not conditional on corporate resources. In columns (3) and (4), the coefficients of the interaction term between *Green* and *Insthold_high* (a dummy variable indicating whether the firm is with a higher than median share of institutional investors or not) are consistently positive and statistically significant, with values of 0.044 and 0.007, respectively. These findings reflect that the negative impact of ESG greenwashing on R&D investment is alleviated by more rigorous monitoring imposed by institutional investors.

Conversely, we observe the negative effect of ESG greenwashing is amplified for firms facing higher market competition and for high-tech firms. In columns (5) and (6), the coefficients of the interaction term between *Green* and *HHI_low* (a dummy variable indicating whether Herfindahl index is lower than median or not) are negative, indicating that greenwashing firms, under increased market competition (or with a low HHI), tend to reduce R&D investment. In columns (7) and (8), the coefficients of the interaction term between *Green* and *Hightech* (a dummy variable indicating whether the firm is high-tech firm or not) are also negative, suggesting that high-tech ESG green-washing firms shirk in R&D investment.

In sum, the results from Table 8 reveal that the negative effect is more pronounced when the firms have less external governance,

Table 6
Robustness check.

Panel A: Subsets of normal firms conditional on the ESG and RRI levels as benchmark. This panel shows the regression results using different subsets of normal firms conditional on the ESG and RRI levels. Columns (1) and (2) identifies brown firms versus normal firms with below-median ESG and versus normal firms with below-median RRI. Columns (3) and (4) identifies green firms versus normal firms with above-median ESG and versus normal firms with above-median RRI. All continuous variables are winsorized at the upper and lower 1 %. See definitions for all variables in .Appendix

	(1)	(2)	(3)	(4)
	Brown versus below-median-ESG Normal	Brown versus below-median-RRI Normal	Green versus above-median-ESG Normal	Green versus above-median-RRI Normal
	$R\&D_{i,t+1}$	$R\&D_{i,t+1}$	$R\&D_{i,t+1}$	$R\&D_{i,t+1}$
$Brown_{i,t}$	0.043***	0.053***		
	(3.324)	(3.791)		
$Green_{i,t}$			−0.015***	−0.027***
			(−3.158)	(−3.168)
Controls	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
N	4,099	4,483	4,238	4,854
Adj. R^2	0.044	0.067	0.174	0.046
F	3.624	5.330	8.815	9.042

Panel B: Alternative outcome variable. This panel shows the regression results using alternative measure for R&D expenditure. R&D′ denotes R&D expenditure (with missing values filled by 0) over total assets. Column (1) uses *Decouple* as the explanatory variable and regress in the full sample; column (2) uses *Brown* as the explanatory variable to identify brown firms versus normal firms, while column (3) uses *Green* to identify green firms versus normal firms. All continuous variables are winsorized at the upper and lower 1%. See definitions for all variables in Appendix.

	(1)	(2)	(3)
	Full sample	Brown versus Normal	Green versus Normal
	$R\&D'_{i,t+1}$	$R\&D'_{i,t+1}$	$R\&D'_{i,t+1}$
$Decouple_{i,t}$	−0.005***		
	(−3.168)		
$Brown_{i,t}$		0.008***	
		(3.132)	
$Green_{i,t}$			−0.003
			(−1.410)
Controls	YES	YES	YES
Industry FE	YES	YES	YES
Year FE	YES	YES	YES
N	8,354	5,772	6,125
Adj. R^2	0.409	0.428	0.441
F	17.488	11.152	20.351

Panel C: Firms change to ESG brown-washing/green-washing. This panel shows the regression results using firm-year observations when the firm changes to ESG brown-washing/green-washing in the next fiscal year. Columns (1) and (2) base on firm-year observations when the firm changes to ESG brown-washing in the next fiscal year (i.e., *Brown* = 0 in *t* and *Brown* = 1 in *t+1*); columns (3) and (4) base on firm-year observations when the firm changes to ESG green-washing in the next fiscal year (i.e., *Green* = 0 in *t* and *Green* = 1 in *t+1*). All continuous variables are winsorized at the upper and lower 1%. See definitions for all variables in Appendix.

	(1)	(2)	(3)	(4)
	$R\&D_{i,t+1}$	$R\&D_{i,t+1}$	$R\&D_{i,t+1}$	$R\&D_{i,t+1}$
$Brown_{i,t}$	0.020*	0.024**		
	(1.893)	(2.160)		
$Green_{i,t}$			0.004	0.004
			(1.125)	(1.048)
$Age_{i,t}$	0.012	0.009	−0.004	0.001
	(0.839)	(0.646)	(−0.389)	(0.090)
$MB_{i,t}$	0.018*	0.019*	0.008***	0.008***
	(1.786)	(1.789)	(2.937)	(2.886)
$Lev_{i,t}$	−0.085*	−0.081**	−0.086***	−0.091***
	(−1.942)	(−2.070)	(−3.702)	(−3.756)
$ROE_{i,t}$	0.003	0.001	−0.002	−0.003
	(0.167)	(0.063)	(−1.203)	(−1.313)
$Growth_{i,t}$	−0.024	−0.019	−0.012	−0.012
	(−0.875)	(−0.742)	(−1.277)	(−1.266)
$Boardsize_{i,t}$		−0.044		−0.012
		(−1.261)		(−0.983)
$Outratio_{i,t}$		0.056		0.021
		(1.094)		(1.285)
$Femratio_{i,t}$		−0.008		−0.009*

(*continued on next page*)

Table 6 (*continued*)

Panel C: Firms change to ESG brown-washing/green-washing. This panel shows the regression results using firm-year observations when the firm changes to ESG brown-washing/green-washing in the next fiscal year. Columns (1) and (2) base on firm-year observations when the firm changes to ESG brown-washing in the next fiscal year (i.e., *Brown* = 0 in *t* and *Brown* = 1 in *t+1*); columns (3) and (4) base on firm-year observations when the firm changes to ESG green-washing in the next fiscal year (i.e., *Green* = 0 in *t* and *Green* = 1 in *t+1*). All continuous variables are winsorized at the upper and lower 1%. See definitions for all variables in Appendix.

	(1)	(2)	(3)	(4)
	$R\&D_{i,t+1}$	$R\&D_{i,t+1}$	$R\&D_{i,t+1}$	$R\&D_{i,t+1}$
		(−0.726)		(−1.683)
$Dual_{i,t}$		0.032		−0.017**
		(1.490)		(−2.002)
$Tenure_{i,t}$		−0.002		−0.001
		(−0.931)		(−1.397)
_cons	0.030	0.158	0.061**	0.108***
	(0.615)	(1.433)	(2.290)	(2.680)
Industry FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
N	1,228	1,228	1,044	1,044
Adj. R^2	0.247	0.257	0.295	0.318
F	1.582	1.226	3.630	3.396

t-statistics with standard errors clustered in the firm level are in parentheses. ***, ** and * indicate statistical significance at the 1 %, 5 %, and 10 % levels, respectively.

Table 7

Cross-sectional regression regarding brown firms versus normal firms: financial constraints and market competition. This table shows the cross-sectional regression results to further identify brown firms versus normal firms in terms of R&D expenditure. Columns (1) and (2) partition firms by the median of *Size*, with *Big_firm* denoting firm of bigger size; columns (3) and (4) partition firms by the median of WW index, with *WW_high* denoting firms with higher financial constraints; columns (5) and (6) partition firms by the median of HHI, with *HHI_low* denoting firms with lower HHI(i.e., subject to more competition in the product market); columns (7) and (8) partition firms by the dummy variable *Hightech* to denote high-tech firms versus non-high-tech firms. All continuous variables are winsorized at the upper and lower 1%. See definitions for all variables in Appendix.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	$R\&D_{i,t+1}$	$R\&D'_{i,t+1}$	$R\&D_{i,t+1}$	$R\&D'_{i,t+1}$	$R\&D_{i,t+1}$	$R\&D'_{i,t+1}$	$R\&D_{i,t+1}$	$R\&D'_{i,t+1}$
$Brown_{i,t}$	0.046***	0.009**	0.014**	0.002	0.037***	0.001	0.039***	−0.000
	(2.775)	(2.561)	(2.232)	(1.051)	(2.861)	(0.390)	(3.382)	(−0.126)
$Brown_{i,t}$*$Big_firm_{i,t}$	**−0.035****	**−0.009****						
	(−2.173)	**(−2.370)**						
$Big_firm_{i,t}$	−0.063***	−0.006**						
	(−4.908)	(−2.185)						
$Brown_{i,t}$*$WW_high_{i,t}$			**0.037****	**0.008****				
			(2.234)	**(2.149)**				
$WW_high_{i,t}$			0.050***	0.002				
			(4.824)	(0.720)				
$Brown_{i,t}$*$HHI_low_{i,t}$					0.023	**0.014*****		
					(0.898)	**(3.101)**		
$HHI_low_{i,t}$					−0.012	−0.008***		
					(−0.662)	(−3.382)		
$Brown_{i,t}$*$Hightech_{i,t}$							0.023	**0.019*****
							(0.785)	**(3.309)**
$Hightech_{i,t}$							0.000	−0.019**
							(0.020)	(−2.579)
Controls	YES	YES	YES	YES	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES	YES	YES
N	5,772	5,772	5,772	5,772	5,772	5,772	5,772	5,772
Adj. R^2	0.068	0.430	0.067	0.429	0.061	0.429	0.061	0.430
F	6.891	9.854	5.785	8.713	3.706	8.680	3.728	9.080

t-statistics with standard errors clustered in the firm level are in parentheses. ***, ** and * indicate statistical significance at the 1 %, 5 %, and 10 % levels, respectively.

face more market competition, and operate in technology-intensive industries. It supports the notion that ESG green-washing firms with decreased R&D investment are subject to managerial opportunism. The managers of ESG green-washing firms are prone to shirking in R&D investment when they are expected to invest in corporate innovation.

Table 8
Cross-sectional regression regarding green firms versus normal firms: firm size and institutional investor holdings. This table shows the cross-sectional regression results to further identify green firms versus normal firms in terms of R&D expenditure. Columns (1) and (2) partition firms by the median of *Size*, with *Big_firm* denoting firm of bigger size; columns (3) and (4) partition firms by the median of institutional investor share-holding, with *Insthold_high* denoting firms with more institutional investor share-holding; columns (5) and (6) partition firms by the median of HHI, with *HHI_low* denoting firms with lower HHI(i.e., subject to more competition in the product market); columns (7) and (8) partition firms by the dummy variable *Hightech* to denote high-tech firms versus non-high-tech firms. All continuous variables are winsorized at the upper and lower 1 %. See definitions for all variables in Appendix.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	$R\&D_{i,t+1}$	$R\&D'_{i,t+1}$	$R\&D_{i,t+1}$	$R\&D'_{i,t+1}$	$R\&D_{i,t+1}$	$R\&D'_{i,t+1}$	$R\&D_{i,t+1}$	$R\&D'_{i,t+1}$
$Green_{i,t}$	−0.008	−0.002	−0.047***	−0.006**	−0.018***	0.004***	−0.021***	0.003***
	(−0.672)	(−0.636)	(−4.057)	(−2.131)	(−2.817)	(2.824)	(−3.392)	(3.337)
$Green_{i,t}$*$Big_firm_{i,t}$	0.003	0.004						
	(0.240)	(0.929)						
$Big_firm_{i,t}$	−0.059***	−0.010***						
	(−5.531)	(−4.147)						
$Green_{i,t}$*$Insthold_high_{i,t}$			**0.044*** **	**0.007***				
			(3.658)	**(1.858)**				
$Insthold_high_{i,t}$			−0.042***	−0.007**				
			(−3.443)	(−2.347)				
$Green_{i,t}$*$HHI_low_{i,t}$					−0.009	**−0.012*** **		
					(−0.889)	**(−3.530)**		
$HHI_low_{i,t}$					−0.004	0.002		
					(−0.396)	(1.122)		
$Green_{i,t}$*$Hightech_{i,t}$							−0.005	**−0.012*** **
							(−0.447)	**(−2.985)**
$Hightech_{i,t}$							0.017	0.001
							(1.518)	(0.293)
Controls	YES	YES	YES	YES	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES	YES	YES
N	6,125	6,125	6,125	6,125	6,125	6,125	6,125	6,125
Adj. R^2	0.075	0.444	0.068	0.442	0.063	0.443	0.063	0.442
F	12.283	15.963	8.537	15.388	8.191	15.573	8.381	15.582

t-statistics with standard errors clustered in the firm level are in parentheses. ***, ** and * indicate statistical significance at the 1 %, 5 %, and 10 % levels, respectively.

5. Conclusion

Our study sheds light on the relationship between ESG decoupling and corporate investment in R&D. We find that ESG decoupling, particularly in the form of green-washing and brown-washing, has distinct effects on firms' subsequent R&D expenditure. When compared to the normal counterparties in the same industry, ESG brown-washing firms tend to increase their R&D investment, while greenwashing firms exhibit a decrease in R&D investment. Our findings remain robust to different subsets of benchmarking normal firms and alternative measurement of the dependent variable. In addition, we find that the capital market only responds positively to ESG green-washing, which implies a favorable attitude toward floated ESG disclosure from the investors.

Furthermore, our study reveals important cross-sectional variations in the relationship between ESG decoupling and R&D investment. Brown-washing firms with limited resources and facing market competition show a stronger positive impact on R&D investment. This indicates that these firms strategically increase R&D spending to upgrade their technology and remain competitive. In contrast, the decrease in R&D investment among green-washing firms is partially mitigated by stringent corporate governance enforced by institutional investors. However, it is further amplified among firms facing greater market competition and operating in high-tech industries, suggesting the presence of managerial opportunism and shirking behavior.

Overall, our research contributes to the understanding of the complex interplay between ESG decoupling and corporate investment in R&D. It highlights the different motivations behind ESG brown-washing and green-washing practices and their implications for corporate innovation input. Our findings underscore the importance of considering ESG decoupling as a factor that can significantly impact firms' strategic decisions and performance. Policymakers and investors should be aware of the potential risks associated with ESG decoupling, while firms need to carefully balance their ESG engagement and innovation investment to ensure long-term competitiveness and sustainable growth.

In terms of the limitations of our study, it is crucial to acknowledge the significance of distinguishing between R&D activities directly associated with ESG and those unrelated to ESG. This distinction greatly impacts our results. Specifically, if R&D is directly linked to the ESG domains, one would expect a mechanistically negative relationship between ESG decoupling and ESG-related R&D investment. This mechanistic relationship indicates that the observed association is not driven by managerial opportunism or strategic incentives. However, due to the constraints of obtaining highly detailed data on R&D expenditures, further exploration of this topic is warranted in future studies.

CRediT authorship contribution statement

Yicheng Sun: Writing – review & editing, Writing – original draft, Data curation, Conceptualization. **Qizhi Tao:** Writing – review & editing, Supervision, Formal analysis. **Du Wang:** Writing – review & editing, Validation, Formal analysis. **Wan Zhang:** Writing – review & editing, Writing – original draft, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix

Variable definitions

Variables	Definition	Source
Outcome variables		
R&D	R&D expenditure (with missing values filled by industry average[1] for each year) over total assets	Compustat
R&D′	R&D expenditure (with missing values filled by 0) over total assets	Compustat
Explanatory variables		
ESG	Bloomberg ESG disclosure score	Bloomberg
RRI	Annual average of Peak_RRI.[2]	RepRisk
Decouple	Equals one if the firm has both above-median ESG and above-median RRI within the three-digit SIC cluster for the year; equals minus one if the firm has both below-median ESG and below-median RRI within the three-digit SIC cluster for the year; zero otherwise.	Bloomberg; RepRisk
Brown	Dummy variable equals one if *Decouple* is minus one, and zero otherwise. A proxy for firms engaging in ESG brown-washing.	Bloomberg; RepRisk
Green	Dummy variable equals one if *Decouple* is one, and zero otherwise. A proxy for firms engaging in ESG green-washing.	Bloomberg; RepRisk
Control variables		
Lev	Leverage	Compustat
ROE	Return on equity	Compustat
Q	Tobin's Q	Compustat
MB	Market to book ratio	Compustat
Age	Firm age	Compustat
Growth	Sale growth	Compustat
Size	Natural logarithm of the firms' total assets plus one.	Compustat
Boardsize	Board size	Boardex
Outratio	The proportion of outside directors in the board	Boardex
Femratio	The proportion of female directors in the board	Boardex
Dual	Dummy variable equals one if CEO is also the chairman of the company, and zero otherwise.	Boardex
Tenure	The number of years that CEO hold the position.	Boardex
Cross-sectional analysis variables		
WW_high	Dummy variable equals one if WW index is above median for the year, and zero otherwise.	Compustat
HHI_low	Dummy variable equals one if the firm lies in the industry with below median Herfindahl index (HHI)[3] for the year, and zero otherwise.	Compustat
Big_firm	Dummy variable equals one if the firm's *Size* is above median for the year, and zero otherwise.	Compustat
Insthold_high	Dummy variable equals one if the firm's institutional investor share-holding is above median for the year, and zero otherwise.	Refinitiv
Hightech	Dummy variable that equals one if the firm is in the "Computer", "Biotech/Medical", "Electronics", and "Telecom" industries which follows Puri and Zarutskie (2012)'s detailed SIC classification, and zero otherwise.	Compustat

[1]The industry cluster is based on three-digit SIC code.

[2]Peak RepRisk Index (RRI) denotes the highest level of ESG risk exposure over the last 24 months. It is a proxy for overall ESG and business conduct risk exposure constructed by RepRisk. As suggested by RepRisk, it should be the main metric used in analyzing the risk exposure of a company

[3]HHI is calculated based on firm sale.

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